

SAI RAHUL BHAT GAADHI RAJU

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PROFESSIONAL SUMMARY

Software developer with a B.Tech in Electronics and Communication Engineering, focused on full stack development using Java, Spring Boot, React, and Python. Experienced in building AI-integrated applications, REST APIs, and database-driven systems, with a solid foundation in data structures, algorithms, and OOP. Comfortable debugging complex systems, picking up new tools quickly, and collaborating effectively within team environments. Skilled in Git/GitHub version control and technical documentation, with a consistent focus on writing clean, functional, well-tested code.

EDUCATION

Anurag University — Bachelor of Technology, Electronics and Communication Engineering

CGPA: 7.95/10 | Expected Graduation: 2027

Jawaharlal Nehru Government Polytechnic (JNGPT), Ramanthapur, India — Diploma in Electronics and Communication Engineering (ECE)

CGPA: 9.17/10 (Graduated with Distinction) | 2021 – 2024

TECHNICAL SKILLS

Programming Languages: Java, Python, C, Verilog HDL

Frontend: React, HTML, CSS, JavaScript (Basic), Bootstrap

Backend: Spring Boot, Spring Security, REST APIs

Database: PostgreSQL, SQL

Developer Tools: Git, GitHub, VS Code, Maven, Vite, MATLAB, Vivado, Cadence Virtuoso, Cisco Packet Tracer

Core CS: Data Structures, Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Other: AI, Machine Learning Basics, Linux Basics, Debugging, Testing, Version Control, Technical Documentation

PROJECTS

Smart Hospital Management System | Java 21, Spring Boot 3, Spring Security, React 18, Vite, PostgreSQL, Bootstrap, Axios, Hugging Face API, Maven

- Engineered a full stack hospital management platform using Java, Spring Boot, React, and PostgreSQL, implementing appointment booking, patient record management, and queue tracking.
- Designed and implemented role-based authentication and authorization for Admin, Doctor, and Patient users with Spring Security, ensuring secure, controlled access across the application.
- Integrated an AI-powered doctor recommendation feature using the Hugging Face Inference API to match patients with relevant specialists based on their needs.
- Built and tested REST APIs for appointment booking, real-time queue monitoring, and token generation, following clean backend architecture and documenting endpoints for maintainability.

RTL Projects | Pipelined ALU, Barrel Shifter, 4-bit ALU — Verilog HDL

- Designed, simulated, and functionally verified a pipelined ALU, barrel shifter, and 4-bit ALU in Verilog HDL, debugging timing and logic issues through waveform analysis.

NOVA OS — AI-Powered Desktop Assistant | Python

- Engineered NOVA OS, a modular Python-based AI desktop assistant, structuring automation features into independent, maintainable components with a scalable software design.
- Debugged and refined assistant behavior across automation workflows to improve reliability and overall user experience.

AetherOS — Universal AI Learning Operating System

- Designed AetherOS, an AI-driven learning platform featuring a productivity dashboard and modular workspace architecture, incorporating structured database planning and core software engineering practices.

FPGA-Based Real-Time Power Quality Anomaly Detection | Python, AI/Edge AI

- Developed the software layer for a real-time power quality anomaly detection system, handling data processing, algorithm development, testing, and technical documentation using Python and AI concepts.

INTERNSHIP

Python Programming Intern | VaultofCodes

- Applied Python programming and logic-building techniques to complete hands-on assignments involving automation and debugging, strengthening core problem-solving skills.

ACHIEVEMENTS

- Selected for the Samsung ISWDP Cohort 8 Fellowship.
- Participated in the India Semiconductor Workforce Development Program (ISWDP).

CERTIFICATIONS

Oracle: Cloud Infrastructure 2025 Certified AI Foundations Associate | **Cisco:** CCNA – Introduction to Networks

Google: Introduction to Generative AI | **NPTEL:** Digital Design with Verilog • VLSI Physical Design with Timing Analysis • VLSI Design Flow: RTL to GDS

Maven Silicon: VLSI SoC Design Overview | **Foundation for QC Innovation (IISc Bengaluru):** Quantum Computing Workshop

LANGUAGES

English, Hindi, Telugu